

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Ship 10+ production AI features end-to-end: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, MCP server integrations, and REST API tooling; 394 commits, 7 deployments. - Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

VectorLift	Semantic Search & RAG Engine	2026
- Built semantic search engine in Python on MS MARCO: BM25 sparse retrieval, dense bi-encoder with in-batch negatives, cross-encoder re-ranker; Kafka-driven Qdrant corpus indexing pipeline. - Scored NDCG@10, MRR@10, MAP; built FastAPI serving with per-stage latency tracking and Streamlit dashboard.		
AlgoPulse	Agentic AI Platform	2025
- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load. - Replaced API polling with Supabase Edge Functions for real-time AI streaming; eliminated state loss under peak traffic.		
LLM Eval Harness	Independent Project	2025
- Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks. - Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap CIs and McNemar tests across 6 metrics.		
KubeServe	Kubernetes LLM Inference Platform	2026
- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing. - Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.		

Languages and Technologies

Languages: Python, TypeScript, JavaScript, Java, SQL, Bash AI/ML & Agentic: Claude/GPT-4o/Gemini APIs, MCP, LangChain, RAG, LLM evaluation, prompt engineering, Temporal.io Backend & Data: FastAPI, Node.js, PostgreSQL, Redis, Kafka, Qdrant, Docker, REST APIs Cloud & DevOps: AWS, GCP, Azure, Kubernetes, KEDA, Terraform, GitHub Actions, Grafana, OpenTelemetry		
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Additional Experience and Awards

IEEE Published: Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees. National Hackathon Winner: Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo. Lead at Google Developers Student Club: Led 4+ tech events (hackathons, conferences) reaching 500+ students. Technical Writing: Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog		
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